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REAL-TIME STUDENT FACIAL EXPRESSION DETECTION TO IDENTIFY STUDENT INTEREST IN LEARNING IN CLASS USING CNN AND YOLO

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ABSTRACT

Understanding students' emotional involvement is essential for optimizing teaching strategies and improving learning outcomes in the classroom. This study employed a Research and Development (R&D) method aimed at developing a system to identify students' learning interest based on real-time facial expression detection. The system consists of three main components: input in the form of facial images captured by a webcam, expression classification processed by a laptop, and output in the form of information indicating the level of student interest. The devices used in testing included a Logitech C310 webcam connected to an MSI Cyborg 15 A12VE-074ID laptop, equipped with an Nvidia RTX 4050 GPU. The algorithm used for the recognition and classification process was a Convolutional Neural Network (CNN) implemented through the YOLO version 11 framework. The dataset consisted of 1,128 student facial expression images classified into two main categories: (1) interested expressions, including happy and surprised, and (2) uninterested expressions, including angry, fearful, sad, and disgusted. These expressions represent basic human emotions. The test results showed that the system achieved an accuracy rate of 86.04% and a precision rate of 89.57%, with an average detection time ranging from 31 to 50 milliseconds. Testing conducted over five days of classroom sessions demonstrated that the system was able to operate stably and accurately using a webcam, even under varying lighting conditions, angles, and facial movements, with an average frame rate of 30–32 FPS.

Keywords: Facial Expression Detection, Learning Interest, Real-Time, Convolutional Neural Network (CNN), YOLOv11

1. Introduction

Facial expressions serve as a fundamental mode of nonverbal communication, conveying complex emotional states and levels of cognitive engagement without verbal articulation. These expressions provide valuable cues for interpreting internal psychological conditions, social intentions, and individual responses to environmental stimuli [1]. With the rapid advancement of Artificial Intelligence (AI) and Machine Learning (ML), the ability to automate and improve facial expression recognition (FER) has attracted considerable attention from both the academic and commercial domains. FER technology has been applied across a wide range of sectors, including healthcare, retail, law enforcement, and education. FER offers unique potential in the educational context. Understanding students' affective states, such as interest, confusion, or frustration, can provide real-time insights that allow educators to dynamically adjust teaching strategies, thereby enhancing student engagement and learning outcomes.

Recent advancements in facial expression recognition (FER) systems have been significantly driven by deep learning methods, particularly Convolutional Neural Networks (CNNs), which have demonstrated strong performance in image classification tasks [2]. Real-time FER applications benefit from the increasing accessibility of powerful computational tools, such as PyTorch, the YOLO framework, and edge devices capable of efficiently processing visual data. The integration of these technologies into educational settings enables the dynamic monitoring of student engagement, thereby supporting more responsive and adaptive teaching strategies [3]. However, despite their promising potential, most FER systems implemented in educational contexts remain retrospective or post-hoc, limiting their practical instructional utility. In modern classrooms, the challenge is not merely to deliver academic content but also to ensure that students remain cognitively and emotionally engaged throughout the learning process. Research indicates that student interest significantly influences academic performance, as it directly impacts attention, persistence, and enthusiasm during learning activities [4]. Student engagement is often expressed nonverbally through facial expressions, making it a valuable source of real-time feedback for educators. Therefore, real-time analysis of facial expressions holds great promise as a tool for enhancing pedagogical practices by enabling immediate adjustments based on students' emotional responses. Nonetheless, tracking engagement in real classroom environments is challenging. Teachers must navigate a wide variety of student behaviors and fluctuating attention levels, making consistent observation of each student's interest impractical. Traditional assessment methods, such as surveys and interviews, are time-consuming and susceptible to bias. To address this issue, real-time FER systems offer a non-intrusive solution by continuously monitoring students' facial cues and identifying patterns that indicate engagement or disengagement [5]. The automation of this

process allows for the reliable collection of real-time data, enabling educators to respond adaptively to students' needs and foster a more personalized and effective learning experience.

In the field of education, Huang et al. [6] evaluated a CNN-based facial expression recognition (FER) system specifically designed to detect emotions on digital learning platforms. Wei et al. [7] further explored CNN ensemble models for online learning, emphasizing the variability of emotional expressions caused by environmental factors such as screen exposure and webcam quality. However, these studies primarily focus on virtual learning scenarios and do not extend to physical classroom environments, where engagement signals are likely to be richer and more nuanced than in virtual environments. Therefore, a notable gap remains in the implementation and validation of real-time FER systems in live classroom settings using hardware-efficient, interpretable, and adaptive models. This study addresses this gap by proposing a real-time CNN-based facial expression recognition system implemented using YOLOv11 and PyTorch on accessible hardware platforms. The system aims to identify students' learning interests by distinguishing between facial expressions that indicate engagement (e.g., happiness and surprise) and disengagement (e.g., sadness, fear, and disgust). Grounded in Ekman's taxonomy of universal facial expressions [8] and further elaborated by Ekman and Friesen [9], the proposed model categorizes facial expressions in real time, enabling immediate feedback between student behavior and instructional responses.

The novelty of this study lies in the real-time implementation of facial expression recognition (FER) within physical classroom settings using low-cost hardware and a robust Convolutional Neural Network (CNN) architecture. Unlike previous approaches, which often relied on retrospective analyses or virtual learning datasets, this study emphasizes ecological validity by embedding the system in live classroom environments. This study also contributes to the field by offering a scalable and practical framework for implementing FER in schools that have limited computational resources. The scope of this study includes system development, validation through accuracy and usability testing, and an analysis of its practical implications for learning design and educational outcomes. By bridging AI technology with classroom pedagogy, this study offers a new pathway toward responsive, data-driven education.

2. Method

a. Research Methods and Design

This study employs a Research and Development (R&D) approach utilizing a prototype-based development model, which enables iterative improvements of a facial expression recognition system designed to detect students' learning interest in real-time classroom environments. The prototype development methodology allows researchers to design, test, and refine systems based on continuous feedback and evaluation in real-world conditions. This framework aligns with the recommendations of [9], who emphasized the importance of adaptive, feedback-driven design in the implementation of emotion recognition systems.

b. Research Background and Participants

This study was conducted at a vocational high school in Takalar, South Sulawesi, Indonesia. The research was conducted over six months, covering both the development and implementation phases. The study involved 30 students aged 16–18 years, selected based on regular attendance and consent for facial data collection. Ethical approval was obtained, and student privacy was ensured according to standard research protocols.

c. System Design

1) Configuration Device Hard

The hardware used in the development of this system included an MSI Cyborg 15 A12VE-074ID laptop equipped with an Intel Core i7 processor, 12GB of RAM, an RTX 4050 GPU, and a 512GB SSD. The camera used was a Logitech C310 HD webcam, capable of recording video at a resolution of 1280×720 with automatic focus. The camera was positioned to provide an unobstructed view of the students' faces, ensuring a clear input for real-time processing.

2) Software and Tools

The software stack was built around PyTorch to implement deep learning and YOLOv11 (You Only Look Once) for object detection. PyTorch enables dynamic computational graph construction and efficient implementation of CNN models, whereas YOLOv11 facilitates real-time detection and classification of facial expressions from live video streams. Song et al. [10] noted the superior performance of hybrid CNN-based models in extracting robust features for facial emotion recognition tasks, which supports the selection of this configuration.

d. Dataset Development and Annotation

A custom dataset comprising 1,200 facial expression images of eleventh-grade TKJ 1 student was developed by capturing photographs of them during classroom activities. The images were captured in real-time by the researcher using an iPhone XS Max under varying lighting conditions and student postures to simulate realistic classroom scenarios. Manual annotation was conducted using Label Studio, referencing Ekman's [8] six basic emotions: happiness, sadness, surprise, fear, anger, and disgust. These emotions were further categorized into two groups: interested (happiness, surprise) and uninterested (sadness, fear, anger, disgust), following the classification framework proposed by R. A. Maulana and Hardiansyah [11]. The dataset was then split into training (70%), validation (20%), and testing (10%) subsets.

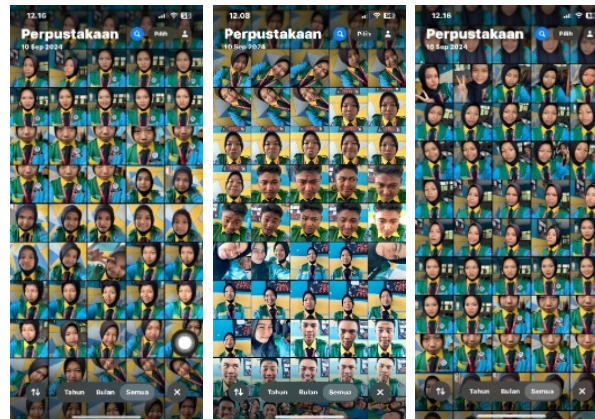


Figure 1. Image dataset expression face

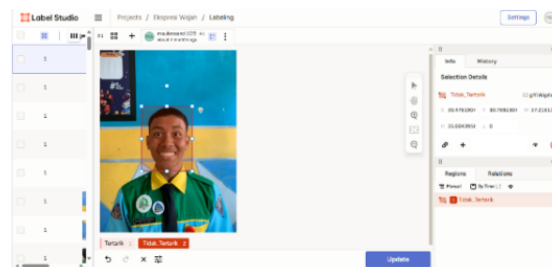


Figure 2. Studio label application

e. CNN Model Architecture and Training

The CNN model architecture consists of multiple convolutional layers for feature extraction, followed by pooling and fully connected dense layers for classification. The input images were normalized to a resolution of 224×224 pixels. The model employs ReLU activation functions in the hidden layers and softmax activation in the output layer for multi-class prediction. Training was conducted using the Adam optimizer with a learning rate of 0.001 and a categorical cross-entropy loss function for 100 epochs. This architectural design aligns with the findings of [12], who emphasized the effectiveness of combining CNNs with attention mechanisms to enhance the reliability of emotion recognition systems.

f. YOLO Integration and Implementation

The You Only Look Once (YOLO) object detection algorithm was integrated into the system to enable real-time facial expression detection and classification. The previously trained, was further refined using the annotated dataset to improve the detection accuracy under classroom conditions. This integration was implemented using Python and OpenCV, allowing for seamless interaction with the facial expression classification module.

During the deployment phase, the model was tested in a classroom environment. YOLO's fast inference capability ensures the efficient processing of video frames, thereby supporting real-time emotion recognition in line with the demands of dynamic and interactive learning settings. In addition, a pre-trained Convolutional Neural Network (CNN) model was incorporated into the YOLOv11 framework to support real-time distribution.

The system was developed using Google Colaboratory and Anaconda Navigator, with a virtual environment named "yolotesis" created to execute Python-based detection scripts. The system initialization involved importing the pre-trained weight file and activating the detection modules. Figures 3 and 4 illustrate the model structure and scripts used to initiate real-time detection.

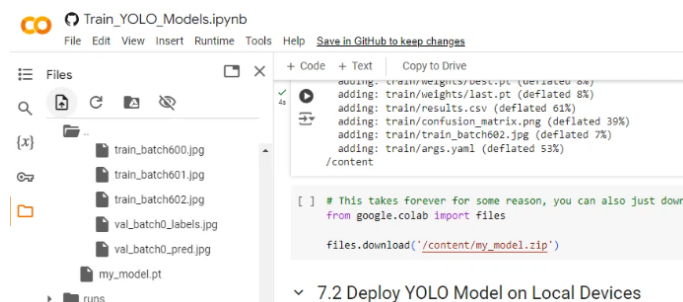


Figure 3. Dataset used to train the model

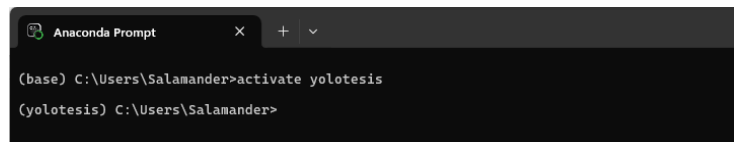


Figure 4. Activating the Yolotesis environment

g. Test Scenario

Observation and scenario-based testing were conducted in alignment with the data collection techniques employed to ensure that the data obtained during the field trials met the research objectives. According to [13], system testing instruments typically involve the use of various customized frameworks and tools, depending on the type of application or system being assessed.

3. Results

A comprehensive testing scenario was designed to evaluate the performance of the integrated facial expression recognition system. The objective of this evaluation was to assess the accuracy, speed, and robustness of the system under various real-world classroom conditions. A diverse group of students from different backgrounds was selected to simulate realistic learning environments. During testing, the participants engaged in typical classroom activities, such as answering questions, participating in discussions, and interacting with peers. The system was assessed based on its ability to detect and classify facial expressions in real time, with a particular focus on its responsiveness to variations in lighting, facial orientation, and emotional intensity. Key performance indicators, such as detection accuracy, processing time per video frame, and overall system reliability, were measured and analyzed. The results yielded valuable insights into the effectiveness of the system and highlighted potential areas for improvement in future iterations.

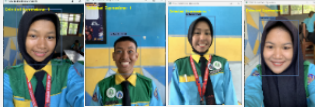
Table 1. Test Scenario

Scenario	Details
1	Similarity in Facial Expression Detection with Dataset
2	Accuracy in Detecting Facial Expressions
3	Multi-Object Detection in Recognizing Facial Expressions
4	Frames Per Second (FPS) in Detecting Facial expressions
5	Realtime System in Detecting Facial expressions

a. Similarity in Facial Expression Detection with Dataset

A similarity test for facial expression detection was conducted using the dataset to evaluate the ability of the system to accurately recognize and classify facial expressions. The primary objective of this test was to assess whether the system could enhance the detection accuracy while minimizing recognition errors. The results of the similarity test based on the dataset are summarized in the following table.

Table 2. Detection with Dataset

Facial Expression Dataset	YOLOv11 Detection System	Information
Happy		In accordance
Sad		In accordance
Surprised		In accordance
Afraid		In accordance

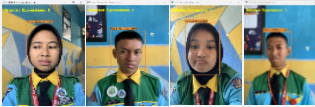
Facial Expression Dataset	YOLOv11 Detection System	Information
Angry		In accordance
Disgusted		In accordance

Table 2 presents the results of the similarity test for facial expression detection, covering six types of expressions: happy, sad, surprised, afraid, angry, and disgusted. In each test, the system successfully recognized the corresponding facial expression with a high degree of accuracy. Specifically, the system accurately identified happy, sad, and surprised expressions in the first three tests. The fourth and fifth tests demonstrated precise recognition of afraid and angry facial expressions, respectively. Finally, the sixth test confirmed the system's ability to detect disgusted expressions effectively. These results indicate that the system performs reliably across a diverse range of facial-expression categories.

b. Accuracy in Detecting Facial Expressions

An accuracy test for facial expression detection was conducted to evaluate the system's ability to recognize various student expressions under different classroom conditions. The test considered several influencing factors, including lighting variations, student seating positions, camera angles, and dynamic changes in facial expressions during the learning process. The results of the accuracy test are presented in the following table.

Table 3. Accuracy in Detecting

Facial Expression Dataset	YOLOv11 Detection System	Information
Accuracy detection	Showing Accuracy in recognize expression face	86.04%
Precision detection	Precision level system in recognize expression face	99.14%

Table 3 presents the results of the accuracy test for facial expression detection, showing an accuracy rate of 86.04%. This indicates that the developed system is effective in recognizing facial expressions and successfully distinguishes between interested and uninterested expressions with minimal error. Testing was conducted over five consecutive school days, with one hour of video recorded each day. Data analysis was performed at 10-minute intervals to ensure consistency and accuracy in recognizing students' expressions. The accuracy rate was calculated using a confusion matrix to evaluate the performance of the classification model.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \times 100\%$$

$$Accuracy = \frac{206 + 22}{206 + 22 + 24 + 13} \times 100\%$$

$$Accuracy = \frac{228}{265} \times 100\%$$

$$Accuracy = 0.86037 \times 100\%$$

$$Accuracy = 86.04\%$$

Note:

Positive True (TP) → Model detects student with Correct as "Interested."

Negative True (TN) → Model detects student with Correct as "Not Interested."

Positive Fake (FP) → Model detects student in a way No Correct as "Interested" even though Actually they are "Not Interested."

Negative Fake (FN) → Model in No appropriate detect student as "Not Interested" even though Actually they are "Interested".

Furthermore, the precision level of facial expression detection was 99.14%. This result indicates that the developed model can accurately distinguish between interested and uninterested expressions with a very low error rate, making it a reliable tool for facial-expression recognition. The precision formula used for this calculation is as follows:

$$\begin{aligned} \text{Precision} &= \frac{TP}{TP + FP} = x 100\% \\ \text{Precision} &= \frac{206 + 22}{206 + 24} = x 100\% \\ \text{Precision} &= \frac{228}{230} = x 100\% \\ \text{Precision} &= 0.99130 x 100\% \\ \text{Precision} &= 99.14\% \end{aligned}$$

Note:

Positive True (TP) → Model detects student with Correct as "Interested."

Positive Fake (FP) → Model detects student in a way No Correct as "Interested " even though they are "Not Interested."

c. Multi-Object Detection in Recognizing Facial Expressions

Multi-object testing for facial expression detection was conducted to evaluate the system's performance in simultaneously recognizing, classifying, and distinguishing various facial expressions. This type of testing is essential for assessing the accuracy of the system in detecting and identifying multiple facial expression instances in real time, its efficiency in handling computational loads, and its reliability in producing consistent results under varying lighting conditions and camera angles. Additionally, the test aimed to examine the system's capability to differentiate between overlapping or visually similar expressions, which is crucial in dynamic classroom environments in which multiple students may concurrently display different emotions simultaneously. The results of the multi-object facial expression detection test are presented in the following table:

Table 4. Multi-Object Detection

Various Face Detection Size	Information
Test 1 (Monday)	Detected
Test 2 (Tuesday)	Detected
Test 3 (Wednesday)	Detected
Test 4 (Thursday)	Detected

Various Face Detection Size		Information
Test 5 (Friday)		Detected
Detecting Faces with Expression Different		Information
Test 1 (Monday)		Detected
Test 2 (Tuesday)		Detected
Test 3 (Wednesday)		Detected
Test 4 (Thursday)		Detected
Test 5 (Friday)		Detected

Table 4 presents the results of the multi-object facial expression detection tests. This evaluation was designed to assess the system's ability to recognize faces of various sizes and accurately detect different facial expressions under various conditions. The test results show that the system effectively adapts to various face sizes and maintains a high classification accuracy across a wide range of emotional expressions. These findings demonstrate that the system has a high level of precision in detecting and identifying facial expressions, confirming its robustness and reliability for practical applications in dynamic, real-world environments.

d. Frames Per Second (FPS) in Detecting Facial expressions

Frames Per Second (FPS) testing was performed to evaluate the speed of the device in detecting facial expressions in real time. Testing This aim was to measure the performance of the device hard in a way as a whole, including speed of data processing, latency of the system, and accuracy of face detection. With the evaluation ability system for operating in real-time, FPS testing provides a valuable outlook on the efficiency and effectiveness of the system in recognizing the expression of students during class activities. Testing this is also important to ensure that the system can adapt to fast dynamics and detect class change expression faces in the right way. FPS test results work as a gauge measuring to evaluate the response system in real time and its suitability for use in a dynamic education environment, where the introduction of fast and accurate face recognition is crucial for supporting interaction learning.

Table 5. Fast and Slow FPS

Number	Exam Day	Fast Action FPS	Slow Motion FPS
Figure A, B	Monday (1)	Speed Per Second: 31.22	Speed Per Second: 20.91
Figure C, D	Tuesday (2)	Speed Per Second: 30.50	Speed Per Second: 20.61
Figure E, F	Wednesday (3)	Speed Per Second: 31.22	Speed Per Second: 22.07
Figure G, H	Thursday (4)	Speed Per Second: 31.99	Speed Per Second: 21.26
Figure I, J	Friday (5)	Speed Per Second: 32.11	Speed Per Second: 20.86

Table 4 presents the results of the multi-object facial expression detection tests. This evaluation was designed to assess the ability of the system to recognize faces of varying sizes and accurately detect different facial expressions under diverse conditions. The results indicate that the system effectively adapts to different face sizes while maintaining a high classification accuracy across a broad spectrum of emotional expressions. These findings demonstrate the high precision of the system in detecting and identifying facial expressions, confirming its robustness and reliability for practical implementation in dynamic, real-world environments.



Figure 5. YOLO FPS system testing

These results indicate that the system demonstrates excellent performance and responsiveness in effectively detecting facial expressions during fast and slow movements. Consequently, the system can operate optimally across various scenarios, ensuring accurate and real-time facial expression detection regardless of movement speed. This ability confirms the resilience and flexibility of the system in adapting to dynamic motion variations, as well as its capacity to maintain high accuracy under unpredictable real-world conditions. Therefore, the system is well-suited for applications that require fast and precise facial expression recognition, such as classroom settings or everyday interactions, where movement variations are common.

e. Realtime System in Detecting Facial Expressions

A real-time facial expression detection test was conducted to evaluate the speed and accuracy of the system in recognizing facial expressions under real-world conditions. The primary objective was to assess the reliability of the system in real-time scenarios, focusing on its ability to accurately identify facial expressions while measuring performance metrics such as latency and frames per second (FPS). Additionally, this test serves as a baseline for optimizing the efficiency of the detection algorithm and identifying potential system weaknesses. By analyzing the real-time performance, this study aims to enhance the system's responsiveness and robustness for practical, real-world applications.

Table 6. Real-time Detection

Motion Detection	Detection YOLOv11 System	Information
Detection Facial Expressions in Real-time Fast Motion		50 ms
Detection Facial Expressions in Real-time Slow Motion		31.25 ms

Table 6 presents the results of the real-time facial expression detection tests. The evaluation under fast-motion conditions demonstrated a detection time of 50 ms per frame, whereas under slow-motion conditions, the detection time was recorded at 31.25 ms per frame. These values indicate the ability of the system to maintain a responsive performance under varying movement speeds. The real-time performance was analyzed using a frame-based formula to calculate the total number of frames that were processed during the test period. This total frame count serves as a critical metric for assessing the effectiveness of the system in maintaining consistent real-time facial expression recognition. A higher frame processing rate reflects the greater efficiency and responsiveness of the system. The formula used to calculate the performance is as follows:

$$\text{Number of Frames} = \frac{FPS}{\text{Videos Duration in Seconds}} \times \text{Total frame}$$

$$\text{Real} = \frac{32}{1 \text{ hour (3600 second)}} \times 115,200 \text{ frameTime Fast Motion}$$

$$\text{Real} = \frac{32}{3600} \times 115,200 \text{ frameTime Fast Motion}$$

$$\text{Real} = \frac{20}{1 \text{ hour (3600 second)}} \times 72,000 \text{ frameTime Slow Motion}$$

$$\text{Real} = \frac{20}{3600} \times 72,000 \text{ frameTime Slow Motion}$$

A per-frame latency analysis was conducted to achieve optimal real-time performance. This analysis measured the time required to process and analyze each video frame from the input stream to accurately identify facial expressions. This step is critical because faster frame processing directly enhances the responsiveness and efficiency of the facial expression detection system. Low latency enables the system to operate with minimal delay, thereby improving the overall user experience and ensuring greater reliability in real-world applications. The formula used to calculate the per-frame latency is as follows:

$$\text{Latency per Frame} = \frac{\text{Number of Frame}}{\text{Total Processing Time (ms)}} = \text{Total Amount (ms)}$$

$$\text{Real} = \frac{115,200}{3600 \times 1000} = 31.25 \text{ ms Time Fast Motion}$$

$$\text{Real} = \frac{115,200}{3600000} = 31.25 \text{ ms Time Fast Motion}$$

$$\text{Real} = \frac{72,000}{3600 \times 1000} = 50 \text{ ms Time Slow Motion}$$

$$\text{Real} = \frac{72,000}{3600000} = 50 \text{ ms Time Slow Motion}$$

Frame latency plays a critical role in the real-time facial expression detection process. Lower latency enables the system to recognize facial expressions more quickly and deliver faster responses, thereby enhancing the detection accuracy and ensuring a smoother user experience. The findings indicate that the developed system and its underlying model can accurately recognize facial expressions in real time. Moreover, the system effectively distinguished between "interested" and "disinterested" expressions with a low error rate, demonstrating both its reliability and efficiency under various environmental conditions. This capability underscores the potential of the system for real-world applications that require responsive and accurate emotion recognition.

4. Discussion

The results demonstrate that the YOLOv11 and CNN-based facial expression recognition system effectively detects and classifies student facial expressions in real time within physical classroom environments. The high accuracy (86.04%) and precision (99.14%) suggest that the system is highly reliable for distinguishing between "interested" and "uninterested" emotional states of the user.

Compared to previous studies on virtual learning environments [3], [6], this study confirms the feasibility of implementing FER technology in live classroom settings, even with low-cost hardware. The consistent performance across various lighting conditions and student positions supports the robustness of the system, aligning with the findings of Tang *et al.* [12] regarding the adaptability of CNNs when combined with optimized detection frameworks.

The system's capability to process 30–32 frames per second (FPS) during fast movements and maintain low latency (<50 ms) ensures real-time responsiveness, which is critical for adaptive learning interventions. This allows educators to receive immediate feedback on student engagement, thereby enabling timely pedagogical adjustment.

However, this study has several limitations. The dataset was limited to a single class group, which may have affected the generalizability of the results. Moreover, while performance remained high under varied conditions, extreme occlusions (e.g., hands covering faces) and very low-light environments still posed detection challenges. Future development could integrate infrared imaging or advanced preprocessing techniques to enhance performance in such cases. Overall, the findings validate that the proposed system offers a practical, scalable solution for the real-time monitoring of student engagement, bridging the gap between AI technology and responsive classroom pedagogy.

5. Conclusion

The student learning interest identification system developed using the YOLOv11 and Convolutional Neural Network (CNN) algorithms has proven effective in detecting students' facial expressions that indicate either interest or lack of interest in real time. The system can distinguish expressions such as happiness and surprise for interested students and sadness, fear, disgust, and anger for those who are uninterested, with a high level of accuracy. Operating at 30–32 frames per second (FPS), the system delivers smooth and responsive facial expression detection in real time. With an overall detection accuracy of 86.04% and a precision rate of 89.57%, the system demonstrated a strong performance in differentiating between interested and uninterested expressions. Testing under various conditions, including changes in lighting, facial position, and student movement, showed that the system remained accurate and reliable even in dynamic classroom environments. The system is also practical for classroom implementation, as it can detect facial expressions within 1–2 s, recognize multiple facial expressions simultaneously in a crowded classroom, operate on low-specification devices, and accurately distinguish students who are engaged from those who are not. The detection results are presented in written form, indicating each student's level of engagement and enabling immediate feedback and interventions.

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